

LIANZHOU, China

WMO: 590720

Lat: 24.8108N

Lon: 112.3719E

Elev: 433

StdP: 14.47

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	34.9	37.1	23.4	17.9	48.0	27.2	21.5	44.5	14.5	49.8	13.1	49.8	3.4	0	0.308

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.6	97.0	78.8	95.2	78.6	93.4	78.3	81.3	90.6	80.6	89.9	79.9	88.6	4.8	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
79.2	153.9	84.3	78.4	149.7	83.6	77.9	147.1	83.2	45.5	90.9	44.7	90.3	44.0	89.0	84.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.5	10.3	8.7	31.0	99.9	2.3	1.9	29.3	101.3	28.0	102.4	26.7	103.4	25.0	104.8
			29.6	82.7	2.5	1.0	27.8	83.4	26.3	84.0	24.9	84.6	23.1	85.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	69.1	49.6	53.8	59.8	69.5	76.8	81.7	84.8	84.2	80.0	72.6	62.9	52.8	
	DBStd	13.76	7.39	9.23	7.86	6.99	4.86	3.91	3.16	3.33	4.84	5.52	6.68	6.03	
	HDD50	209	100	58	11	0	0	0	0	0	0	0	3	37	
	HDD65	1550	480	330	200	37	0	0	0	0	0	9	114	380	
	CDD50	7188	87	164	315	585	831	951	1079	1059	901	701	390	125	
	CDD65	3053	2	16	39	171	366	501	614	594	451	245	51	3	
	CDH74	29660	7	94	195	1024	2895	5126	7386	6742	4255	1633	293	10	
	CDH80	12203	0	12	26	268	1008	2034	3505	3055	1739	503	52	1	
Wind	WSAvg	3.2	3.3	3.4	3.0	3.0	3.0	3.4	3.1	3.2	3.3	3.1	3.2		
Precipitation	PrecAvg	62.1	2.4	3.6	5.5	9.6	11.0	8.5	5.4	5.8	3.4	3.0	2.2	1.6	
	PrecMax	89.8	6.1	11.2	14.6	22.9	20.3	22.5	12.2	16.2	11.5	9.0	7.2	6.2	
	PrecMin	36.6	0.1	0.4	0.7	3.6	1.9	2.4	1.7	0.6	0.0	0.0	0.0	0.1	
	PrecStd	13.1	1.9	2.7	3.0	4.1	4.3	4.5	2.9	3.5	2.5	2.3	2.0	1.5	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	74.4	82.0	83.2	89.7	94.4	96.8	99.7	99.3	96.7	92.3	85.6	74.7	
		MCWB	63.1	67.4	69.6	74.3	77.7	79.4	77.9	79.0	78.4	75.4	70.9	63.9	
	2%	DB	69.2	76.8	79.3	86.5	91.4	94.3	97.1	96.9	93.9	88.3	80.9	69.8	
		MCWB	60.2	65.4	67.4	73.3	75.8	78.9	78.7	79.0	77.5	72.2	66.7	59.6	
	5%	DB	65.2	72.4	75.4	83.1	88.8	92.2	95.2	94.8	91.8	85.5	77.4	66.3	
		MCWB	57.4	63.2	66.3	71.8	75.9	78.6	78.7	78.6	77.1	70.8	65.3	56.7	
	10%	DB	61.2	67.8	71.7	79.9	85.8	89.8	93.0	92.5	89.3	82.6	73.8	62.8	
		MCWB	54.8	61.5	64.9	71.1	75.1	78.5	78.8	78.5	76.4	69.8	63.7	55.3	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	65.5	69.3	73.2	77.3	80.9	82.6	82.3	81.9	80.7	77.7	73.5	68.0
			MCDB	71.5	78.7	79.2	85.7	89.0	90.1	91.8	93.0	90.8	87.0	79.9	72.0
2%		WB	62.1	66.6	70.7	75.4	79.1	81.0	81.1	81.0	79.7	76.0	70.9	63.0	
		MCDB	67.3	74.5	75.6	82.7	87.1	88.8	90.7	91.4	88.9	83.4	77.4	67.0	
5%		WB	58.8	64.3	68.1	74.1	77.9	80.1	80.4	80.2	78.8	74.4	68.7	59.4	
		MCDB	63.8	70.9	72.9	80.4	85.1	87.7	89.9	89.8	87.5	80.6	73.8	63.7	
10%		WB	55.7	61.8	66.0	72.7	76.8	79.3	79.8	79.5	78.0	72.8	66.2	56.2	
		MCDB	60.2	67.6	70.6	78.1	83.0	86.5	88.5	88.1	85.6	78.9	71.7	61.2	
Mean Daily Temperature Range	5% DB	MDBR	11.7	12.0	11.5	12.3	13.1	12.5	14.6	14.8	14.8	15.5	15.4	14.6	
		MCDBR	21.3	21.2	20.3	18.8	19.1	17.3	18.7	19.0	18.9	20.6	22.5	22.4	
		MCWBR	11.8	10.6	9.0	7.2	6.3	4.4	4.0	4.2	4.8	6.3	8.9	11.7	
	5% WB	MCDBR	17.1	17.7	14.7	14.2	14.0	13.7	15.3	15.9	16.4	14.7	15.6	15.8	
		MCWBR	10.5	9.6	8.8	7.1	6.0	4.7	4.5	4.6	4.8	6.5	8.3	10.0	
Clear-Sky Solar Irradiance	taub	0.667	0.721	0.834	0.803	0.658	0.616	0.591	0.663	0.704	0.728	0.626	0.630		
	taud	1.569	1.513	1.372	1.453	1.746	1.852	1.910	1.719	1.622	1.534	1.691	1.652		
	Ebn at Noon	187	189	178	189	219	227	232	216	202	185	196	189		
	Edh at Noon	78	88	106	99	74	66	62	75	81	84	68	69		
All-Sky Solar Radiation	RadAvg	663	694	764	916	1128	1230	1529	1439	1296	1158	913	796		
	RadStd	153	152	132	141	160	181	156	106	115	143	151	178		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+103

Nomenclature: See separate page